

Hands-On-Learning: Mastering Sequential Agent Workflows

Description

In this lab, you will solve a practical challenge in content marketing by designing and implementing a sequential multi-agent workflow. You will use Flowise AI to chain together three specialized agents that collaborate to automatically generate a blog post from a single topic, demonstrating your understanding of task decomposition and communication protocols.

Learning Objectives

After completing this lab, you will be able to:

- Decompose a complex task (content creation) into a logical sequence of sub-tasks.
- Implement a sequential workflow in Flowise where the output of one agent becomes the input for the next.
- Engineer specific prompts to guide each agent's role within the collaborative chain.
- Analyze the benefits and limitations of a linear communication protocol in a multi-agent system.



Assignment Components

Introductory Slide:

- **Title:** Assembly Line AI: Mastering Sequential Agent Workflows
- **Subtitle:** Decompose a complex task and build a collaborative pipeline with chained AI agents.

Tools to be used:

- Flowise AI (web-based or locally installed version)
- A text editor (e.g., Google Docs, MS Word, Notepad) for drafting prompts.

Scenario

You are an AI Workflow Specialist at "ContentCraft Marketing," an agency struggling to keep up with content demands. The current process of writing a blog post is slow and manual. Your manager has asked you to create a prototype for an automated content generation pipeline.

The goal is to build a system where a single topic (e.g., "The Benefits of Remote Work") can be fed into a workflow that automatically produces a research outline, a draft article, and SEO suggestions. This requires creating a "digital assembly line" where specialist agents perform their task and then pass their work to the next agent in the sequence.

Instructions for learners

Part 1: Building the Agent Chain

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Create a new Chatflow. Log into Flowise and start with a blank canvas. This canvas will contain your entire agent pipeline.

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Build Agent 1: The "Researcher." Add an LLM Chain node. Configure its prompt to accept a (topic) and generate a bulleted list of key points or a structured outline for an article on that topic.

Example Prompt Snippet: "You are a world-class researcher. Given the topic '{topic}', generate a 5-point bulleted outline for a blog post..."

Build Agent 2: The "Writer." Add a second LLM Chain node. This agent's job is to write a draft article. Crucially, its prompt must be designed to use the output from the "Researcher" agent.

3 Tip: In the prompt for the "Writer," you will use the variable that holds the output of the first chain.

Example Prompt Snippet: "You are a skilled content writer. Using the following outline: write a short, engaging 3-paragraph blog post..."

Build Agent 3: The "SEO Analyst." Add a third LLM Chain node. This final agent will analyze the article draft from the "Writer" and provide SEO feedback.

4 Example Prompt Snippet: "You are an SEO expert. Based on the article provided: '{draft_from_writer}', suggest 3 SEO-friendly title options and 5 relevant keywords."

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Create the Sequential Chain. Go to Chains > Simple Sequential Chain. This special node will be your pipeline controller. Connect your three LLM Chain nodes to it in the correct order: Researcher -> Writer -> SEO Analyst.

6

Connect and Test. Connect a final Chat Input node to your Simple Sequential Chain. Now, test the entire workflow by entering a topic like "The Future of AI in Marketing." Verify that the final output includes the SEO suggestions based on the article written from the research outline.

Part 2: Documentation & Analysis

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Prepare your deliverables. Once your pipeline is working, gather the required information (screenshot, prompts, analysis) as specified in the section below.

Deliverables for Submission (AI Grading)

1 Deliverable 1: Final Workflow Canvas

Question Prompt: "Upload a single, full-screen screenshot of your complete Flowise canvas. The screenshot must clearly show the three LLM Chain nodes (Researcher, Writer, SEO Analyst) correctly connected within a Simple Sequential Chain."

Response Type: File Upload

File Types: PNG, JPG, JPEG

2 Deliverable 2: Agent Prompt Engineering

- Question Prompt: "Submit a document containing the exact text of the three prompts you engineered for your 'Researcher,' 'Writer,' and 'SEO Analyst' agents. For each prompt, add one sentence explaining how it effectively transforms its specific input to produce a structured output for the next agent in the chain."
- Response Type: Rich Text or File Upload
- File Types: PDF, DOCX, TXT

3 Deliverable 3: Communication Protocol Analysis

Question Prompt: "In 150-200 words, analyze the 'Simple Sequential Chain' you built as a form of a communication protocol. Explain one major advantage and one significant limitation of this linear, one-way information flow for multi-agent collaboration. Reflect on the process of breaking down the 'content creation' task. What was the most critical consideration when designing the prompts to ensure a smooth handoff of information between your agents?"

Response Type: Rich Text, txt, pdf or docx

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.